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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Integration of DUAL-BATTERY Architecture in the DYNAMIC WIRELESS POWER TRANSFER SYSTEMS(DWPT)

Team ID: *GEN26-A0H-I060*

Team Name: *DynaCharge*

Domain: *2*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

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Problem Statement

Battery-powered systems have limited operating time and require frequent charging, leading to operational downtime and reduced productivity. As the demand for continuous operation increases, maintaining reliable and uninterrupted power availability remains a major challenge.

The Specific DWPT Challenge

During dynamic wireless power transfer (DWPT), received power and load demand can vary as the platform moves. If charging input and load current act on the same battery, current fluctuations and internal losses may increase heating and battery stress, depending on operating conditions. Similar to a mobile phone becoming warm when used while charging, the effect can be more demanding in high-power systems and may contribute to faster ageing.

Problem Understanding

- **Who is affected?** EV users and fleet operators, warehouse and factory operators using AMRs/AGVs, and other users who depend on electrically powered equipment for long operating shifts.
- **Why does it occur?** Batteries have finite capacity; conventional charging can require downtime, while DWPT power varies with coil alignment, air gap, speed, and transmitter availability.
- **Thermal stress:** Charging current, load current, internal resistance, conversion losses, and changing demand can contribute to heat generation. The actual effect depends on battery chemistry, current, temperature, and control conditions.
- **Battery ageing:** Repeated high-current or high-temperature operation can accelerate capacity fade and increase internal resistance, potentially reducing useful service life.
- **Operational impact:** Unplanned stops, reduced availability, energy losses, maintenance costs, and uncertainty in reliable runtime.
- **Safety concern:** Poorly managed temperature, voltage, or current can compromise reliability and may create unsafe operating conditions.

Existing Gaps & Challenges

- **Battery heating in DWPT:** Dynamic changes in received power, load, and conversion efficiency can affect battery temperature.
- **Potential battery degradation:** Repeated thermal and electrical stress may reduce capacity and useful life. The impact of charging input while the battery also supplies a load needs to be quantified under defined operating conditions.
- **Preset switching thresholds:** Some dual-battery strategies rely on fixed SOC or voltage thresholds, which may not respond well to changing temperature, load, or DWPT availability.
- **Integration gap:** In our initial literature exploration, we found limited directly relevant material combining dual-battery switching, DWPT, and thermal-aware control in one validated framework. This is a gap to investigate, not a claim that no related work exists.

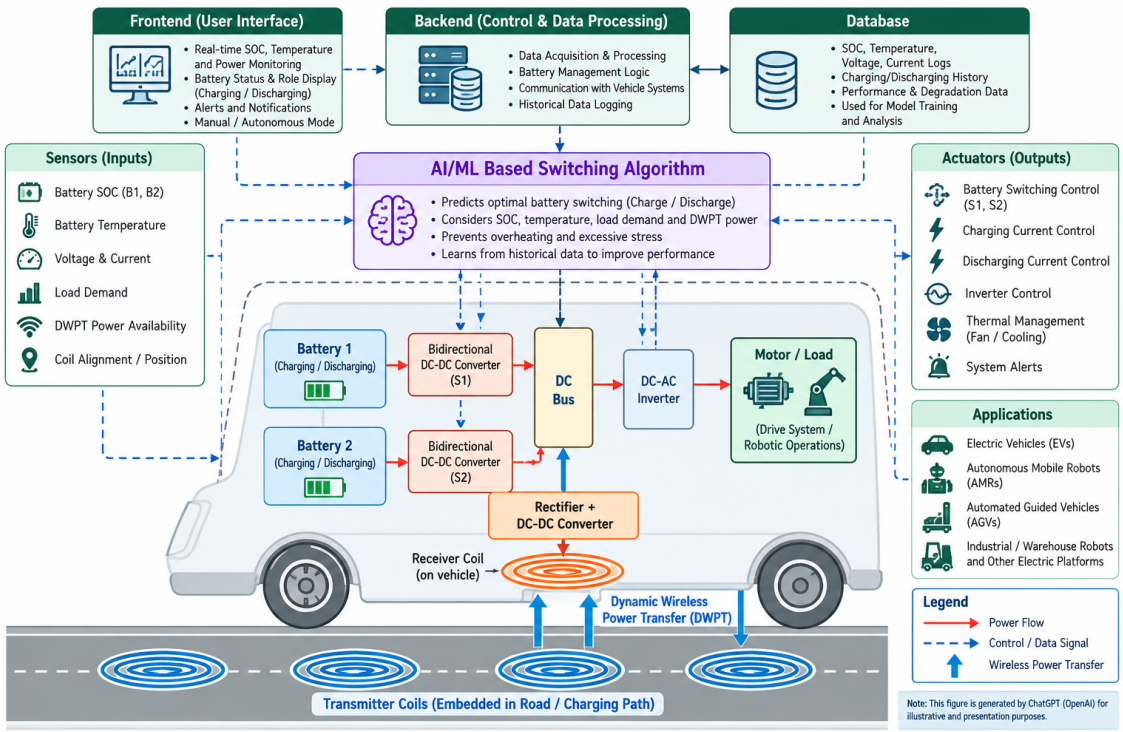
Main Unanswered Challenge

How can battery heating and ageing be reduced when charging power and load demand vary during DWPT, without compromising output power, safety, or operating time? A phone warming during use while charging is a familiar analogy; high-power systems require their own measurements and models.

Our Proposed Solution

- **Core idea:** Integrate dynamic wireless power transfer (DWPT) with two independently managed battery units for electrically powered platforms.
- **Coordinated operation:** When conditions permit, one battery supplies the load while the other receives wireless charging. Their roles can switch when required.
- **Adaptive decisions:** Consider SOC, temperature, voltage/current, load demand, and available DWPT power instead of relying only on fixed thresholds.
- **Safety and power conversion:** Use appropriate switching and DC–DC conversion with current/voltage limits, temperature safeguards, and controlled transitions.
- **Development plan:** Compare the concept in MATLAB/Simulink first, then pursue hardware validation. EVs, AMRs, and AGVs are target applications; longer battery life and improved efficiency are outcomes to measure, not assume.

Technical Approach



Architecture illustration generated with ChatGPT (OpenAI AI); conceptual, not a validated hardware schematic.

Frontend
Planned dashboard for SOC, temperature, current, battery role, and alerts.

Backend
MATLAB/Simulink for system simulation; Python may be used for data processing and control prototyping.

Database
No production database established. CSV or SQLite can store simulation logs and test data.

AI/ML switching logic
Evaluate SOC, temperature, load demand, and DWPT availability; compare adaptive logic with fixed thresholds.

Hardware
Battery units, sensors, switching circuits, converters, and transmitter/receiver coils for future validation.

Innovation & Uniqueness

- **Integrated concept:** Combines dual-battery energy management, DWPT, temperature monitoring, and adaptive switching within one system architecture.
- **Thermal-aware decisions:** Switching decisions consider SOC and temperature together with load demand and wireless power availability, rather than SOC alone.
- **Broader application range:** The architecture is intended to be adaptable to EVs, AMRs, AGVs, and other electrically powered platforms at suitable power levels.
- **Additional protection:** The design considers voltage/current regulation, safe switching transitions, thermal limits, and possible regenerative-energy integration.
- **Validation focus:** Compare against a single-battery and/or fixed-threshold baseline using temperature, current peaks, energy efficiency, output stability, downtime, and battery-ageing indicators. These benefits require testing.

Feasibility & Scalability

- **Literature exploration:** The team used Scopus AI and Leapspace AI to explore related research and assess feasibility. The initial search covered DWPT, dual-battery systems, battery management, thermal effects, and power converters; the specific reports should be cited in the project records.
- **Feasibility:** Begin with a MATLAB/Simulink model and compare single-battery, fixed-threshold dual-battery, and adaptive thermal-aware operation under ideal and varying-load/DWPT conditions.
- **Key constraints:** Reliable battery thermal models, sensor accuracy, coil alignment and coupling, converter losses, switching transients, battery protection, and safe experimental power levels.
- **Scalability:** The control concept may be adapted to EVs, warehouse AMRs, AGVs, and industrial equipment, with power electronics and battery sizing tailored to each application.
- **Next validation step:** Establish measurable baselines before claiming longer battery life or greater DWPT range; proceed to a low-power prototype only after simulation and safety review.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1 – Electric Vehicles (EVs):** Vehicles travelling through DWPT-enabled road segments could receive energy while moving, with coordinated battery operation intended to reduce charging interruptions and manage thermal stress.
- **Use case 2 – Autonomous Mobile Robots (AMRs):** Warehouse robots transporting goods between storage, picking, and packing areas could use charging zones along their routes to reduce time spent out of service.
- **Use case 3 – Automated Guided Vehicles (AGVs):** Factory and logistics AGVs moving materials between workstations could use embedded or designated wireless charging paths to support longer shifts and reduce charging-related stoppages.

Impacts and Benefits

- **Daily users and operators:** Potentially fewer charging-related interruptions, improved equipment availability, and more predictable operation for EV fleets, warehouse robots, and industrial vehicles.
- **Long-term benefit:** If thermal and electrical stress are reduced, batteries may retain useful capacity for longer, potentially lowering replacement and maintenance costs.
- **Environmental impact:** Longer battery service life could reduce the frequency of battery replacement and the associated battery waste and resource demand.
- **Expected measurable outcomes:** Compare battery temperature, current peaks, energy efficiency, output-power stability, downtime, and capacity/ageing indicators against a single-battery baseline under matched conditions.
- **Range objective:** Evaluate whether DWPT integration improves usable operating distance or operating time; any increase must be demonstrated through testing rather than assumed.

Sustainable Development Goals (SDGs)

SDG	Target	Goal Name	Connection to DynaCharge
SDG 7	7.3	Affordable and Clean Energy	Improved energy efficiency is a project objective to be evaluated through simulation and testing.
SDG 9	9.4	Industry, Innovation and Infrastructure	Supports exploration of more efficient and sustainable energy infrastructure for EVs and industrial mobile platforms.
SDG 12	12.5	Responsible Consumption and Production	If battery service life increases, battery replacement frequency and related waste may be reduced.

Individual Member Contribution

Member	Contribution or responsibility
Sathi Siva Ramakrishna Reddy	Team leader; researched dual-battery systems and DWPT; proposed their integration; contributed to the initial chart design; gathered literature and feasibility insights using Scopus AI and Leapspace AI.
Galanagari Karthik Reddy	Reviewed two DWPT research papers and contributed to selecting and refining the content for Chart 1.
Reshma	Designed the chart drawings and written content; reviewed research on coil alignment and related DWPT factors.
Maanasha S	Reviewed a dual-battery research paper, identified relevant stakeholders, and contributed to chart design.

Yukti ID

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